MODULE 3: FUNCTIONAL MATERIALS
(CO: 2)

Complete Study Textbook, Step-by-Step Derivations & VIT FAT Question Bank

Aligned with Official VIT BACHY101 Course Page Syllabus, Question Bank, and Past FAT Papers

Scope of This Textbook (Official VIT Syllabus - All 10 Topics): (1) Electrical materials: Ohm's law and resistivity, (2) Classification: metals and semiconductors, (3) Intrinsic and extrinsic semiconductors, superconductors, (4) Magnetic materials: para and dia, (5) Ferromagnetic materials, (6) Antiferromagnetism and applications in storage devices, (7) Liquid cooling: types of coolants, thermal conductivity, applications, (8) Nanomaterials introduction, (9) Top-down and bottom-up synthesis, and (10) Bulk vs. nano (gold), quantum dots.

1 VIT FAT Exam Blueprint & Master Formula Sheet

Module 3 carries **15 to 20 marks** in BACHY101 FAT papers:

1. **Question Type A (10 Marks):** Band Theory of solids (metals, semiconductors, insulators) and derivation of conductivity $\sigma = ne\mu_e + pe\mu_h$ for intrinsic vs extrinsic semiconductors.
2. **Question Type B (10 Marks):** Magnetic classification (dia, para, ferro, antiferro, ferri) with spin alignment diagrams, hysteresis curve (coercivity, remanence), and hard vs soft magnetic storage devices.
3. **Question Type C (10 Marks):** Nanotechnology: Top-down vs bottom-up synthesis, quantum confinement in quantum dots, and surface plasmon resonance in nano gold vs bulk gold.

Concept	Governing Equation / Relation	VIT Exam Key Insight
Ohm's Law & Conductivity	$J = \sigma E, \quad \sigma = \frac{1}{\rho} = ne\mu$	Conductivity decreases with T in metals; increases exponentially with T in semiconductors.
Intrinsic Carrier Concentration	$n_i = p_i = \sqrt{N_c N_v} e^{-\frac{E_g}{2k_B T}}$	Fermi level lies exactly at mid-gap: $E_F = \frac{E_c + E_v}{2}$.
Curie-Weiss Law (Ferro)	$\chi = \frac{C}{T - \theta} \quad (T > T_C)$	Above Curie temperature T_C, ferromagnets become paramagnetic.
Meissner Effect (Superconductors)	$B = \mu_0(H + M) = 0 \Rightarrow \chi = -1$	Superconductors exhibit perfect diamagnetism below T_c.
Thermal Conductivity	$q = -kA\frac{dT}{dx}$ (Fourier's Law)	Nanofluids suspend metal/oxide nanoparticles to boost heat transfer by 40%.
Brus Equation (Quantum Dots)	$E_g^{\text{nano}} = E_g^{\text{bulk}} + \frac{h^2}{8m^* R^2} - \frac{1.8e^2}{4\pi\varepsilon R}$	Bandgap widens as nanoparticle radius R decreases (blue shift).

2 Electrical & Semiconductor Materials

2.1 Band Theory of Solids & Ohm's Law

In isolated atoms, electrons occupy discrete atomic orbitals. In crystalline solids, atomic orbitals overlap to form continuous energy bands:

- **Valence Band (VB):** Highest occupied band containing valence electrons.
- **Conduction Band (CB):** Lowest empty or partially filled band.
- **Forbidden Bandgap (E_g):** Energy separation between VB maximum and CB minimum.

1. **Metals (Conductors):** VB and CB overlap ($E_g = 0$) or CB is partially filled. Resistivity increases linearly with temperature due to increased electron-phonon scattering.
2. **Semiconductors:** Moderate bandgap ($E_g \approx 0.7$ to 2.0 eV). At 0 K, they are insulators; at room temperature, thermal excitation excites electrons across E_g.
3. **Insulators:** Extremely wide bandgap ($E_g > 5.0$ eV).

2.2 Extrinsic Semiconductors & Superconductors

- **n-type Semiconductor:** Silicon (Group IV) doped with pentavalent donors (P, As, Sb). Extra electron occupies donor energy level E_d right below E_c. Majority carriers: electrons ($n \approx N_d$).
- **p-type Semiconductor:** Silicon doped with trivalent acceptors (B, Al, Ga). Acceptor level E_a lies right above E_v. Majority carriers: holes ($p \approx N_a$).
- **Superconductivity:** Phenomenon where electrical resistance vanishes completely ($\rho = 0$) below a critical temperature T_c. In the superconducting state, magnetic field lines are completely expelled from the bulk interior (**Meissner Effect**).

3 Magnetic Materials & Data Storage

3.1 Classification by Magnetic Dipole Ordering

1. **Diamagnetism:** Atoms possess no permanent magnetic dipole moment. When exposed to external magnetic field H, induced magnetic moments oppose H ($\chi < 0$, small and temperature independent). Examples: $\mathrm{Bi}, \mathrm{Cu}, H_2O$.
2. **Paramagnetism:** Permanent atomic magnetic moments exist due to unpaired electrons, but are randomly oriented by thermal agitation. External H aligns dipoles ($\chi > 0$, obeys Curie's Law: $\chi = \frac{C}{T}$). Examples: $\mathrm{Al}, O_2, \mathrm{Ti}$.
3. **Ferromagnetism:** Permanent dipoles exhibit spontaneous parallel alignment within localized microscopic regions called **magnetic domains** via quantum mechanical exchange interaction. Large positive susceptibility ($\chi \gg 1$). Examples: $\mathrm{Fe}, \mathrm{Co}, \mathrm{Ni}$.
4. **Antiferromagnetism:** Adjacent magnetic dipoles align anti-parallel with equal magnitude, resulting in zero net spontaneous magnetization below Neel temperature T_N. Example: $\mathrm{MnO}, \mathrm{FeO}$.
5. **Ferrimagnetism:** Adjacent dipoles align anti-parallel with unequal magnitudes, yielding net spontaneous magnetization. Examples: Ferrites ($\mathrm{Fe_3O_4}, \mathrm{NiFe_2O_4}$) used in high-frequency transformer cores and microwave circulators.

3.2 Hysteresis Loop & Magnetic Recording

When a ferromagnet is cycled through a magnetic field:

- **Remanence (B_r):** Residual magnetic induction remaining when H is reduced to zero.
- **Coercivity (H_c):** Reverse magnetic field required to demagnetize the material to zero.
- **Hard Magnetic Materials:** High coercivity and high remanence with broad hysteresis loops (e.g., Alnico, NdFeB). Used in permanent magnets and magnetic hard disk drive (HDD) recording platters.
- **Soft Magnetic Materials:** Low coercivity and narrow hysteresis loops with minimal hysteresis energy loss (e.g., Silicon steel, Permalloy). Used in transformer cores and motor stators.

4 Nanomaterials & Quantum Confinement

4.1 Synthesis: Top-Down vs. Bottom-Up

- **Top-Down Approaches:** Bulk macroscopic material is progressively reduced down to nanoscale dimensions via mechanical or chemical etching (e.g., High-energy ball milling, photolithography, electron beam lithography).
- **Bottom-Up Approaches:** Nanostructures are built atom-by-atom or molecule-by-molecule via controlled chemical assembly (e.g., Sol-gel processing, chemical vapor deposition [CVD], hydrothermal synthesis).

4.2 Bulk vs. Nano Gold & Quantum Dots

- **Surface-to-Volume Ratio:** As particle diameter drops from 100 nm to 2 nm, the proportion of surface atoms surges from 1% to over 50%, drastically lowering the melting point and boosting catalytic activity.

- **Surface Plasmon Resonance (SPR):** Bulk gold is shiny yellow and chemically inert. Gold nanoparticles ($10 - 30$ nm) exhibit collective oscillation of conduction electrons in resonance with visible light, absorbing green-blue photons and displaying intense ruby-red color (used in biosensors and lateral flow assays).
- **Quantum Dots:** Semiconductor nanocrystals ($2 - 10$ nm) exhibiting 3D quantum confinement. Because particle size is smaller than the exciton Bohr radius, continuous energy bands collapse into discrete atom-like energy levels. The emission wavelength can be precisely tuned across the entire spectrum by tuning crystal size.